

Section 2: Structures and functions in living organisms

- a) Levels of organisation
- b) Cell structure
- c) Biological molecules
- d) Movement of substances into and out of cells
- e) Nutrition
- f) Respiration
- g) Gas exchange
- h) Transport
- i) Excretion
- j) Coordination and response

a) Levels of organisation

Students will be assessed on their ability to:

- 2.1 describe the levels of organisation within organisms: organelles, cells, tissues, organs and systems.

b) Cell structure

Students will be assessed on their ability to:

- 2.2 describe cell structures, including the nucleus, cytoplasm, cell membrane, cell wall, chloroplast and vacuole
- 2.3 describe the functions of the nucleus, cytoplasm, cell membrane, cell wall, chloroplast and vacuole
- 2.4 compare the structures of plant and animal cells.

c) Biological molecules

Students will be assessed on their ability to:

- 2.5 identify the chemical elements present in carbohydrates, proteins and lipids (fats and oils)
- 2.6 describe the structure of carbohydrates, proteins and lipids as large molecules made up from smaller basic units: starch and glycogen from simple sugar; protein from amino acids; lipid from fatty acids and glycerol
- 2.7 describe the tests for glucose and starch
- 2.8 understand the role of enzymes as biological catalysts in metabolic reactions
- 2.9 understand how the functioning of enzymes can be affected by changes in temperature, including changes due to change in active site
- 2.10 understand how the functioning of enzymes can be affected by changes in active site caused by changes in pH**
- 2.11 describe experiments to investigate how enzyme activity can be affected by changes in temperature.

d) Movement of substances into and out of cells

Students will be assessed on their ability to:

- 2.12 understand definitions of diffusion, osmosis and active transport
- 2.13 understand that movement of substances into and out of cells can be by diffusion, osmosis and active transport
- 2.14 understand the importance in plants of turgid cells as a means of support**
- 2.15 understand the factors that affect the rate of movement of substances into and out of cells, to include the effects of surface area to volume ratio, temperature and concentration gradient
- 2.16 describe experiments to investigate diffusion and osmosis using living and non-living systems.

e) Nutrition

Students will be assessed on their ability to:

Flowering plants

- 2.17 describe the process of photosynthesis and understand its importance in the conversion of light energy to chemical energy
- 2.18 write the word equation and the balanced chemical symbol equation for photosynthesis
- 2.19 understand how varying carbon dioxide concentration, light intensity and temperature affect the rate of photosynthesis
- 2.20 describe the structure of the leaf and explain how it is adapted for photosynthesis
- 2.21 understand that plants require mineral ions for growth and that magnesium ions are needed for chlorophyll and nitrate ions are needed for amino acids
- 2.22 describe experiments to investigate photosynthesis, showing the evolution of oxygen from a water plant, the production of starch and the requirements of light, carbon dioxide and chlorophyll

Humans

- 2.23 understand that a balanced diet should include appropriate proportions of carbohydrate, protein, lipid, vitamins, minerals, water and dietary fibre**
- 2.24 identify sources and describe functions of carbohydrate, protein, lipid (fats and oils), vitamins A, C and D, and the mineral ions calcium and iron, water and dietary fibre as components of the diet
- 2.25 understand that energy requirements vary with activity levels, age and pregnancy**
- 2.26 describe the structures of the human alimentary canal and describe the functions of the mouth, oesophagus, stomach, small intestine, large intestine and pancreas
- 2.27 understand the processes of ingestion, digestion, absorption, assimilation and egestion
- 2.28 explain how and why food is moved through the gut by peristalsis
- 2.29 understand the role of digestive enzymes, to include the digestion of starch to glucose by amylase and maltase, the digestion of proteins to amino acids by proteases and the digestion of lipids to fatty acids and glycerol by lipases
- 2.30 understand that bile is produced by the liver and stored in the gall bladder, and understand the role of bile in neutralising stomach acid and emulsifying lipids
- 2.31 describe the structure of a villus and explain how this helps absorption of the products of digestion in the small intestine
- 2.32 describe an experiment to investigate the energy content in a food sample.**

f) Respiration

Students will be assessed on their ability to:

- 2.33 understand that the process of respiration releases energy in living organisms
- 2.34 describe the differences between aerobic and anaerobic respiration
- 2.35 write the word equation and the balanced chemical symbol equation for aerobic respiration in living organisms
- 2.36 write the word equation for anaerobic respiration in plants and in animals
- 2.37 describe experiments to investigate the evolution of carbon dioxide and heat from respiring seeds or other suitable living organisms.**

g) Gas exchange

Students will be assessed on their ability to:

- 2.38 understand the role of diffusion in gas exchange

Flowering plants

- 2.39 understand gas exchange (of carbon dioxide and oxygen) in relation to respiration and photosynthesis
- 2.40 understand that respiration continues during the day and night, but that the net exchange of carbon dioxide and oxygen depends on the intensity of light**
- 2.41 explain how the structure of the leaf is adapted for gas exchange
- 2.42 describe the role of stomata in gas exchange
- 2.43 describe experiments to investigate the effect of light on net gas exchange from a leaf, using hydrogen-carbonate indicator**

Humans

- 2.44 describe the structure of the thorax, including the ribs, intercostal muscles, diaphragm, trachea, bronchi, bronchioles, alveoli and pleural membranes
- 2.45 understand the role of the intercostal muscles and the diaphragm in ventilation
- 2.46 explain how alveoli are adapted for gas exchange by diffusion between air in the lungs and blood in capillaries
- 2.47 understand the biological consequences of smoking in relation to the lungs and the circulatory system, including coronary heart disease
- 2.48 describe experiments to investigate the effect of exercise on breathing in humans.

h) Transport

Students will be assessed on their ability to:

- 2.49 understand why simple, unicellular organisms can rely on diffusion for movement of substances in and out of the cell
- 2.50 understand the need for a transport system in multicellular organisms

Flowering plants

- 2.51 describe the role of phloem in transporting sucrose and amino acids between the leaves and other parts of the plant**
- 2.52 describe the role of xylem in transporting water and mineral salts from the roots to other parts of the plant
- 2.53 explain how water is absorbed by root hair cells
- 2.54 understand that transpiration is the evaporation of water from the surface of a plant
- 2.55 explain how the rate of transpiration is affected by changes in humidity, wind speed, temperature and light intensity
- 2.56 describe experiments to investigate the role of environmental factors in determining the rate of transpiration from a leafy shoot

Humans

- 2.57 describe the composition of the blood: red blood cells, white blood cells, platelets and plasma
- 2.58 understand the role of plasma in the transport of carbon dioxide, digested food, urea, hormones and heat energy
- 2.59 explain how adaptations of red blood cells, including shape, structure and the presence of haemoglobin, make them suitable for the transport of oxygen
- 2.60 describe how the immune system responds to disease using white blood cells, illustrated by phagocytes ingesting pathogens and lymphocytes releasing antibodies specific to the pathogen
- 2.61 understand that vaccination results in the manufacture of memory cells, which enable future antibody production to the pathogen to occur sooner, faster and in greater quantity**
- 2.62 understand that platelets are involved in blood clotting, which prevents blood loss and the entry of micro-organisms**
- 2.63 describe the structure of the heart and how it functions
- 2.64 explain how the heart rate changes during exercise and under the influence of adrenaline
- 2.65 describe the structure of arteries, veins and capillaries and understand their roles
- 2.66 understand the general structure of the circulation system to include the blood vessels to and from the heart, the lungs, the liver and the kidneys.

i) Excretion

Students will be assessed on their ability to:

Flowering plants

- 2.67 understand the origin of carbon dioxide and oxygen as waste products of metabolism and their loss from the stomata of a leaf

Humans

- 2.68 recall that the lungs, kidneys and skin are organs of excretion
- 2.69 understand how the kidney carries out its roles of excretion and osmoregulation
- 2.70 describe the structure of the urinary system, including the kidneys, ureters, bladder and urethra
- 2.71 describe the structure of a nephron, to include Bowman's capsule and glomerulus, convoluted tubules, loop of Henlé and collecting duct
- 2.72 describe ultrafiltration in the Bowman's capsule and the composition of the glomerular filtrate
- 2.73 understand that water is reabsorbed into the blood from the collecting duct
- 2.74 understand that selective reabsorption of glucose occurs at the proximal convoluted tubule
- 2.75 describe the role of ADH in regulating the water content of the blood
- 2.76 understand that urine contains water, urea and salts.

j) Coordination and response

Students will be assessed on their ability to:

- 2.77 understand that organisms are able to respond to changes in their environment
- 2.78 understand that homeostasis is the maintenance of a constant internal environment and that body water content and body temperature are both examples of homeostasis
- 2.79 understand that a coordinated response requires a stimulus, a receptor and an effector

Flowering plants

- 2.80 understand that plants respond to stimuli
- 2.81 describe the geotropic responses of roots and stems
- 2.82 describe positive phototropism of stems

Humans

- 2.83 describe how responses can be controlled by nervous or by hormonal communication and understand the differences between the two systems
- 2.84 understand that the central nervous system consists of the brain and spinal cord and is linked to sense organs by nerves
- 2.85 understand that stimulation of receptors in the sense organs sends electrical impulses along nerves into and out of the central nervous system, resulting in rapid responses
- 2.86 describe the structure and functioning of a simple reflex arc illustrated by the withdrawal of a finger from a hot object
- 2.87 describe the structure and function of the eye as a receptor
- 2.88 understand the function of the eye in focusing near and distant objects, and in responding to changes in light intensity**
- 2.89 describe the role of the skin in temperature regulation, with reference to sweating, vasoconstriction and vasodilation**
- 2.90 understand the sources, roles and effects of the following hormones: ADH, adrenaline, insulin, testosterone, progesterone and oestrogen.